Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam C · Form 2: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. Two identical Atwood machines have the same mass difference. In trial B an identical ballast is added to each pan. Which comparison is correct?

(A) B has less acceleration at the same driving force

(B) B has more acceleration because both pans are heavier

(C) B has the same acceleration because tension is equal

(D) B accelerates in the opposite direction

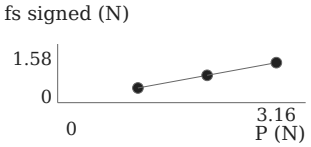
2. An incline block of weight 10 N rests on a 30° ramp with a 5 N hanger on an ideal pulley. Add vertical downward force P to the block. Friction is sufficient; uphill is positive. P (N) -> fs signed (N): (1.16, 0.58); (2.16, 1.08); (3.16, 1.58). Which horizontal setting produces a reading of 0.83?

(A) 2.988

(B) 0.996

(C) 1.66

(D) 2.324



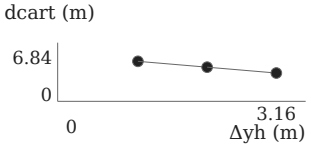
3. A cart and a hanging mass share a taut rope over one ideal fixed pulley. Measure downward hanger displacement and the cart distance from the pulley. Delta yh (m) -> dcart (m): (1.16, 6.84); (2.16, 5.84); (3.16, 4.84). Set the horizontal origin at the first measurement. What is the new vertical intercept?

(A) 4.104

(B) 12.31

(C) 6.84

(D) 9.576



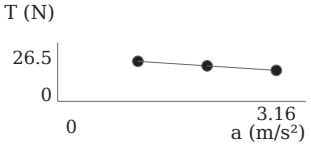
4. An unseen hanger rides on one side of an ideal pulley. Vary the opposite load and record tension with the unseen hanger acceleration positive downward. Use g=10 m/s². a (m/s²) -> T (N): (1.16, 26.52); (2.16, 23.52); (3.16, 20.52). By what factor does the first-to-third change exceed the first-to-second change?

(A) 2

(B) 3.6

(C) 1.2

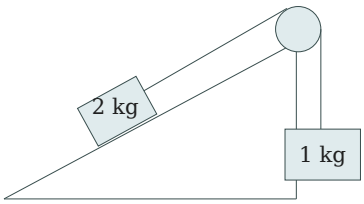
(D) 2.8



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A 2 kg block on a smooth 30° incline shares an ideal rope with a 1 kg hanger. Use g=10 m/s².

a. Show the block weight components in incline coordinates. **(1 pt)**



b. Explain the motion and predict the direction after a small increase in ramp angle. **(3 pt)**

6. Release a 60 kg cabin and adjustable counterweight over an ideal pulley without a motor. Cabin up is positive; g=10 m/s². acabin (m/s²) -> Tcable (N): (1.16, 669.6); (2.16, 729.6); (3.16, 789.6).

a. Build the model for counterweight elevator from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. A cart of mass M on a smooth track and hanger of mass m are connected over an ideal pulley. Predict the tension change if the cart mass doubles and the hanger is unchanged. **(3 pt)**